Abdul Rub Mohammed

Data Scientist | https://www.linkedin.com/in/abdulrubmohammed/
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PROFESSIONAL SUMMARY

Data Scientist with 3+ years at Infosys building ML models, statistical analyses, and Python/SQL/Spark pipelines for enterprise insurance clients; M.S. Data Science candidate at UNT (GPA 3.9, Dec 2026). Turns ambiguous business questions into measurable models and stakeholder-ready insights, cutting insurance document processing from ~1 hour to minutes and reducing stakeholder report corrections by 30%.

SKILLS

- **Statistics & ML:** Regression & Time-Series Analysis, Hypothesis Testing, Cross-Validation, Hyperparameter Tuning, Scikit-learn, TensorFlow, Keras, PyTorch, LSTM/RNN.
- **Generative AI & LLMs:** RAG, LangChain, LLM Evaluation, Prompt Engineering, Fine-Tuning, Groq API, Anthropic Claude API.
- **Languages & Data:** Python, SQL, Java, C++, Spark, Hive, Hadoop, Pandas, NumPy, statsmodels.
- **Cloud & MLOps:** AWS (S3, SageMaker, DynamoDB), Azure ML, Azure OpenAI, GCP, Docker, Git.
- **Analytics & Visualization:** Pandas, NumPy, Tableau, Power BI, Looker, Matplotlib, statsmodels.

EXPERIENCE

Optiu.ai | AI Engineer Intern ***June 2026 – Aug 2026***

- Co-developed LLM-driven sales-forecasting and customer-retention agents for SMBs in the carwash and grocery sectors, combining time-series regression on client sales data with Groq-hosted LLMs to turn forecasts into plain-language recommendations; piloted with 4 prospective clients (2 per sector).
- Developed Python integrations with the Groq API and automated tests for agentic workflows (optimization, deal-closing) in the OptiU Platform SDK, enabling rapid iteration across agent releases.

University of North Texas | Math Lab Tutor ***Feb 2026 - Present***

- Tutor students one-on-one and in groups in probability & statistics, linear algebra, and calculus; over 80% of tutored students improved their course grades.
- Translate complex quantitative concepts into accessible frameworks, strengthening the analytical foundation directly applicable to ML model development and statistical inference to the students.

Infosys | Associate Data Scientist ***2021 - 2024***

- Co-designed the evaluation framework for a RAG-based document co-pilot, validating extraction accuracy and error rates against a ground-truth set of several thousand enterprise documents to ensure reliability before production rollout.
- Developed an ML-assisted insurance document-processing pipeline on AWS (Python, scikit-learn, DynamoDB), cutting processing time from ~1 hour to a few minutes per batch for downstream claims and underwriting teams.
- Streamlined stakeholder reporting with Tableau dashboards on AWS-backed Python and built-in data-quality checks, cutting report production from 1–2 hours to minutes per batch and reducing stakeholder correction requests by 30%.
- Automated generation of outgoing insurance correspondence (claim letters and return/refund notices to insureds and policy owners) with Java, saving 8–10 minutes per document across 200+ weekly transactions (~30 staff-hours/week).
- Engineered Spark/Hive and SQL pipelines that fed unstructured enterprise documents into the co-pilot and integrated validated outputs with SAP via Java workflows.
- Earned 2 Infosys INSTA Awards as one of 5 engineers delivering AI solutions for international insurance clients, partnering remotely with on-site client leads in Agile sprints and shipping 2 weeks ahead of schedule.

PROJECTS

- **Healthcare Claims Outcomes Analysis**
  Analyzed 1,000+ healthcare claims to profile outcomes, denial reasons, and payment patterns; identified the leading denial reasons and their share of total denials and surfaced disparities in denial rates to inform revenue-cycle prioritization. **Tech: Python, Tableau, Pandas.**
- **Scalable Ride-Sharing Database System** https://github.com/MohammedAbdulRub/rideshare_platform
  Designed a normalized relational database schema with constraint-based business rules to optimize end-to-end ride-sharing operations such as booking lifecycle, secure payments, and feedback using analytical SQL queries supporting 10K+ simulated ride transactions. **Tech: SQL, MySQL, RDBMS, Constraint Modeling.**
- **Human Activity Recognition using LSTM**
  Trained a 2-layer stacked LSTM on raw UCI HAR smartphone sensor data, reaching 90.8% mean test accuracy (93.1% best) versus 86.8–88.2% for CNN baselines; an accelerometer-only ablation dropped accuracy to 87%. **Tech: Python, TensorFlow/Keras, Scikit-Learn.**

EDUCATION

- **University of North Texas** Master of Science | Data Science | GPA: 3.9 ***Jan 2025 – Dec 2026***
- **Osmania University** Bachelor of Engineering | Computer Science & Engineering